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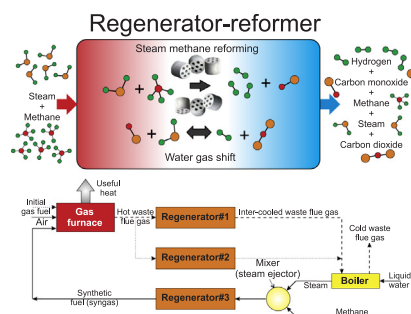
Numerical study of steam methane reforming over a pre-heated Ni-based catalyst with detailed fluid dynamics

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GRAPHICAL ABSTRACT



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ABSTRACT

A computational fluid dynamics (CFD) model of steam methane reforming process over a pre-heated Ni-based catalyst is developed via ANSYS Fluent for real computational domain of the reformer. The investigations are focused on determination of temperature and mole fraction of individual element distribution in the reformer at various initial conditions. The fluid dynamics of reacting flow is studied in experimental setup. It is established that the increasing of the gas flow rate leads to pressure loss increasing. For example, at velocity inlet of 1 m/s the pressure drop is 512 Pa for the investigated reformer. The results of CFD simulation are revealed that the mole fractions of each elements are constantly changing due to chemical reactions when the reaction mixture passes through catalyst bed of the steam methane reformer. When the synthesis gas composition obtained with the CFD model versus the equilibrium syngas composition is compared, it is presented that at the catalyst temperature above 1300 K a syngas composition is close to equilibrium. The distribution of the gas temperature in the reformer is determined by the developed model. The temperature gradient near the reformer inlet has a maximum value, because the temperature drop between catalyst and reaction mixture is maximum and maximum rate of steam methane reforming reaction takes place in this section of the reformer. The temperature gradient slows down to outlet of reformer, because the catalyst temperature decreases due to endothermic steam methane reforming reaction, and also due to increase of steam-to-methane ratio from inlet to outlet due to methane consumption (for inlet steam-to-methane ratio above unit). For follow initial conditions (steam-to-methane ratio is 2.0, pressure is 5 bar, residence time is $4.54 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$, temperature of feed stock is 800 K, catalyst temperature is 1300 K), temperature of outlet synthesis gas is 849 K.

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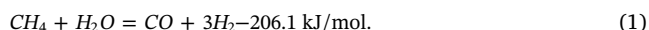
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1. Introduction

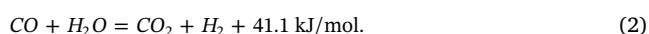
In accordance with the hydrogen scenario of the European Commission, large-scale hydrogen production in the world will begin after 2030 and will be stimulated by a significant reduction in the cost of hydrogen technologies and increased consumption of hydrogen in the transport sector [1]. Nowadays the major technology for production of hydrogen and synthesis gas (mixture of CO (carbon monoxide) and H₂ (hydrogen)) is steam reforming of hydrocarbon: methane [2–5], ethanol [6–8], methanol [9–11], biomass [12–14] and other hydrocarbons [15–17]. The production of hydrogen by natural gas reforming accounts for about 48% of the world's hydrogen production [18]. The main component of natural gas is methane; therefore steam methane reforming is the major way for hydrogen production.

The chemical reaction of steam methane reforming (SMR) is well studied. SMR process is comprised of heating the reaction mixture (steam and methane) over a various type of catalysts [2–4,19,20]. According previous investigations it can be concluded that the overall steam methane reforming process produces hydrogen, carbon monoxide, carbon dioxide and minute quantity of other chemicals. But only three reactions have substantial contribution to heat and mass balance [2,21–23]:

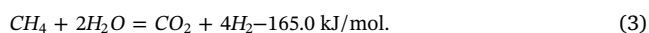
–steam methane reforming reaction:



–water-gas shift reaction:



–global steam methane reforming reaction:



Generally, the industrial reaction of steam methane reforming is carried out in reformers – large-scale furnace [24–27]. In the reformer, steam methane reforming reactions are normally taken place at moderate high temperatures 900 to 1200 K in a presence of catalysts [21]. In the industry, various types of catalysts for steam methane reforming are well characterized. Usually, Ni-based catalysts are used in industrial scale [21,28–30]. In addition, other catalysts, for example noble metals-bases, are also applied for steam methane reforming [31,32]. A large number of scientific publications and investigations concerning the study of steam reforming of methane and devices for its implementation allow to constantly improving SMR process.

Recently, numerical modeling and investigation has been actively used to study steam methane reforming. Hoang and co-workers developed 2D model of steam methane reformer and compared the obtained results with 1D model and experimental data [21]. In that study the chemical reactions over the catalyst bed, mass and heat transfers in the radial and axial directions in the steam methane reformer are considered. The optimal conditions for high energy efficiency of hydrogen production in steam methane reforming process (temperature range of 700–800 °C under steam-to-methane ratio 3–3.5) was reported by Hoang and co-authors [21]. These obtained results also are confirmed by other investigations [33–35].

Sunol and Patel presented a steady-state model of thermally coupled steam methane reformer that is consisted of three ducts [36]. Combustion of natural gas occurs in the first duct which is activated by noble metal-based catalyst and steam methane reforming process carry out in the second duct which is activated by nickel-based (Ni/Al₂O₃) catalyst. The developed model made it possible to determine the temperatures of reaction mixture and mole fractions of the reacting components in the reformer.

With considerable increase of computers computational power, CFD-simulation has become an progressively increasing way for steam methane reformer development, integrating chemical (chemical reactions) and physical (heat-and-mass transfer) models for different geometry of SMR-reformer. The main advantage of CFD-modeling is a powerful visualization of the calculation results for subsequent processing and analysis. In addition, computational fluid dynamics modeling provides versatility to modify design parameters without the expense of research equipment changes which brings large money economic and economy of time [37,38]. The CFD-modeling results of various combined chemical and physical phenomena are in well agreement with the experimental data for different reformer geometries, types of catalysts and operation conditions [39,40,21]. Various commercial and open-source software are used for modeling and simulation of steam methane reforming process, for example, ANSYS Fluent [41,42], Comsol Multiphysics [43], Autodesk Simulation CFD [44], OpenFOAM [45] (open source). Lao et al. [39] presented a computational fluid dynamics (CFD) model of an large-scale SMR process for hydrogen production. A developed CFD-model is implemented in ANSYS Fluent for real computational domain to predict temperatures and mole fractions of each element into reformer. The simulation results can be used to improve the energy efficiency of SMR process, and also can be applied to develop efficient reactors.

Dixon [40] presented a comparison of the results obtained via CFD-modeling of steam methane reforming process in Comsol Multiphysics. Author simulated the chemical reactions and temperature distribution for 1D, 2D and 3D reactor models. The most accurate data, which are in good agreement with the experimental results, are obtained for the 3D and 2D model. Also, Ku and co-workers [46] described comprehensive CFD-discrete element method of numerical model for the reactor which is implemented in open-source software OpenFOAM.

CFD-modeling studies on steam methane reforming process (mentioned above) were mainly focused on investigation of reformer, in which heat for endothermic SMR reaction supplied through the wall [39,40,21]. This paper is devoted to CFD investigation of steam methane reforming process over pre-heated Ni-based catalyst. The catalyst of reformer bed, heated by hot flue gases, is cooled through taking place a strongly endothermic steam methane reforming reaction. This method of steam methane reforming is adopted for hydrogen and synthesis gas production and also can be used for thermochemical waste-heat recovery [47–51]. Therefore, 2D CFD model of steam methane reforming process over pre-heated Ni-based catalyst is developed in ANSYS Fluent to predict the reaction and transport phenomena into reaction space of reformer. The major intention is paid to the investigation of mole fraction and temperature distribution into reformer bed for various operation conditions. In addition, much attention is paid to studying the fluid dynamics of the reacting flow, because when the reacting components pass through the catalyst bed, there are pressure losses. The results of steam methane reforming study which is implemented by Froment and Xu [2] is used for receiving equations of chemical conversion rate over a Ni-based catalyst.

2. Schematic diagram and experimental setup

2.1. Schematic diagram

Steam methane reforming over a preheated catalyst can be used not only for hydrogen and synthesis gas production. This method of SMR can be used for thermochemical waste-heat recuperation [48,49,52]. This method of improving of energy efficiency is based on the principle of transformation of the flue gases heat into the chemical energy of a synthetic fuel, which is consisting mainly of hydrogen.

Fig. 1 shows a schematic diagram of steam methane reforming process over a preheated Ni-based catalyst for thermochemical waste-heat recuperation system. A recuperation system of methane-consuming industrial furnace is consisted of at least three regenerator (steam

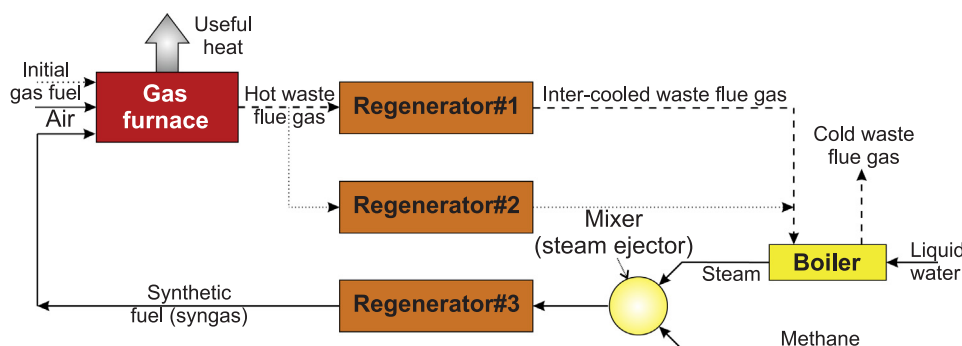


Fig. 1. The schematic diagram of steam methane reforming process over a preheated Ni-based catalyst.

methane reformer) for waste-heat recuperation. Steam after the boiler is mixed with methane in a gas mixer (steam ejector). A mixture of methane and steam is fed to a reformer, the catalyst bed of which is preheated by hot flue gases. The reaction of steam methane reforming is carrying out on a preheated catalyst, which produces a synthesis gas. Reformate, consisting mainly of hydrogen and carbon monoxide, can be used as fuel for the furnace or for other purposes [53,54].

Fig. 2 shows the graphical representation of catalyst preheat and cooling cycles of steam methane reforming process. Whereas a 1st and 2nd regenerator-reformers are being preheated by heat of hot waste flue gas after combustion chamber of furnace, a 3rd regenerator-reformer, which is preheated throughout the previous cycles, is cooled by taking place the steam methane reforming process that is strongly endothermic chemical reaction. When the third regenerator-reformer is cooled by the steam methane reforming process, the hot waste flue gas is redirected to this reformer while the second regenerator, which is now hot, is employed for taking place the steam methane reforming process. The same sequence is with the second and third reformers. Subsequently the cooling and preheat cycles are repeated.

2.2. Experimental setup

Many CFD investigations of steam methane reforming mainly focus on the study of the temperature and mole fraction of chemical elements distribution in the reaction space of the reformer [39,40,21]. However, these works paid little attention to the study of the pressure loss of the reaction mixture passing through the reformer. In order to investigate the fluid dynamics and the pressure loss of the reaction mixture, the catalyst bed, presented on Fig. 3, was studied by experiment and CFD modeling.

Mechanical and thermal characteristics of investigated catalyst are presented in Table 1. Values of mechanical characteristics (composition, density, surface area) were provided by the catalyst manufacturer NIAP-CATALYZATOR LLC (Novomoskovsk, Russia). The specific heat was obtained by author at experimental setup, which is described

below.

The catalyst structure that obtained by SEM (Scanning Electron Microscopy) is presented in Fig. 4.

2.3. Experimental setup

To study the fluid dynamics, pressure losses and thermal properties of the catalyst the experimental setup shown in Fig. 5 is used. The experimental setup mainly consists of the following elements: 1 – electric heater for heating the gas flow to 600°; 2 – Pitot tube for measuring static and dynamic pressure; 3 – a cylindrical tube that is filled with a catalyst; 4 – instrument board presenting the pressure and flow temperature.

The flowmeter for measuring gas flow rate is not installed on the experimental setup. Therefore the gas flow rate is determined by the following expression:

$$Q = \frac{\pi \cdot d^2}{4} \cdot \sqrt{\frac{2 \cdot p_{dyn}}{0.95 \cdot \rho}}, \quad (4)$$

where d – diameter of the cylindrical tube, m; p_{dyn} – dynamic pressure, Pa; ρ – gas density, kg/m³; 0.95 – recovery factor of Pitot tube [55].

3. Numerical setup

A two dimensional steam methane reformer model is built and numerically resolved via Ansys Fluent CFD code. The computational domain of the reformer model presented Fig. 6.

The phenomenon accounted in the steam methane reformer embrace chemical and physical process such as endothermic reactions on the catalyst bed, heat transfer between catalyst and reaction flow and gas diffusion in the axial and radial directions of the reformer.

The SMR reformer is simulated as the pseudo-homogenous reactor. The following assumptions were made for the developed model: – pseudo-homogenous model of the reformer; – constant catalyst layer porosity in x (axial) and y (radial) directions; – reformer wall is

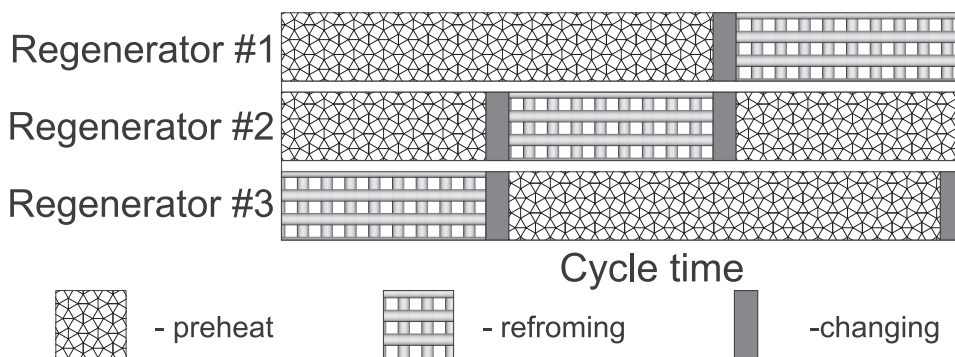


Fig. 2. The graphical representation of catalyst preheat and cooling cycles.

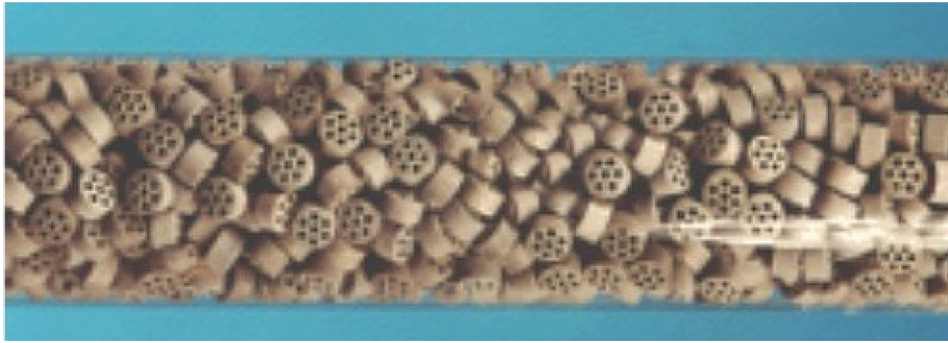
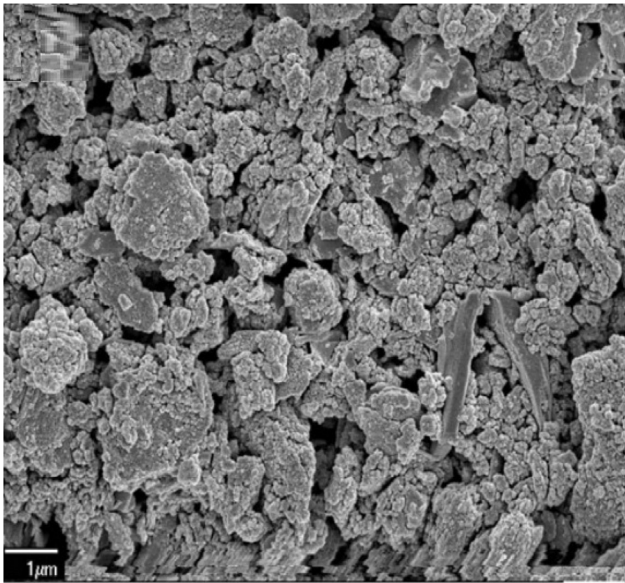


Fig. 3. Reformer filled with catalyst.

Table 1

Technical characteristics of Ni-based catalyst.

Characteristic	Value
Active component: NiO	14.5%
Active component: SiO ₂	0.2%
Catalyst support	CaO-MgO-La ₂ O ₃ -Al ₂ O ₃
Bulk density	680 kg/m ³
Surface area	3.0–5.0 m ² /g _{cat}
Thermal Conductivity	18.2 W/(m·K)
Specific Heat	880 J/(kg·K)

Fig. 4. SEM images for Ni/Al₂O₃ catalyst.

thermally perfect; – heat loss to surroundings is neglected; – reaction flow is ideal gases; – steam methane reforming is steady-flow process.

3.1. Governing equation

The continuity equation, written in a general form:

$$\frac{\partial \rho}{\partial \tau} + \nabla \cdot (\rho \cdot u) = 0, \quad (5)$$

where u – gas velocity, m/s; ρ – gas flow density, kg/m³.

Taking into account chemical reactions and diffusion of gases in the reformer, the continuity Eq. (5) are written as follows [21]:

$$u \cdot \frac{\partial C_i}{\partial x} = D_i \left(\frac{\partial^2 C_i}{\partial y^2} + \frac{1}{y} \frac{\partial C_i}{\partial y} + \rho_{cat} \cdot r_i \right), \quad (6)$$



Fig. 5. Experimental setup for investigation fluid dynamics, pressure losses and thermal properties of the catalyst.

where x and y – axial and radial direction, respectively; C_i – mole fraction of i -reaction element; D_i – diffusion coefficient of i element; ρ_{cat} – catalyst density, kg/m³.

Conservation of momentum equation [56]:

$$\rho \left(u_x \frac{\partial u_x}{\partial x} + \frac{\partial u_x}{\partial y} \right) = -\frac{\partial P}{\partial x} + \mu \nabla^2 u_x + f_{turb,x}; \quad (7)$$

$$\rho \left(u_y \frac{\partial u_y}{\partial x} + \frac{\partial u_y}{\partial y} \right) = -\frac{\partial P}{\partial y} + \mu \nabla^2 u_y + f_{turb,y}, \quad (8)$$

where u_x and u_y – axial and radial components of gas velocity u , m/s; P – pressure, Pa; f_{turb} – external body force; μ – dynamic viscosity, Pa·s.

Ideal gas equation [56]:

$$P = \rho R_0 T \sum \frac{m_i}{M_i}, \quad (9)$$

where R_0 – universal gas constant, J/(K·mol); T – temperature, K; m_i and M_i – mass and molar mass of i component.

Energy equation for reaction space of the reformer [21,36]:

$$-\rho_{cat} \cdot c_{p(cat)} \frac{\partial T}{\partial \tau} + \rho_{cat} \sum (\Delta H_j) \cdot \eta_j \cdot R_j + S_h \cdot h_{cat} (T_{cat} - T_g) + K \left(\frac{\partial^2 T}{\partial y^2} + \frac{1}{y} \frac{\partial T}{\partial y} + \frac{\partial^2 T}{\partial x^2} \right) = 0, \quad (10)$$

where $c_{p(cat)}$ – specific heat of catalyst, J/(kg·K); ΔH_j – enthalpy of j -reaction, J/mol; S_h – surface area to catalyst layer volume, m²/m³; h_{cat} – heat transfer coefficient between hot catalyst and reaction gas flow, W/(m²·K); T_{cat} , T_g – temperature of catalyst bed and gas flow, respectively, K; K – thermal conductivity of catalyst bed, (W/(m·K)); η_j – effectiveness factor of j -reaction; R_j – rate of j reaction, mol/(s·kg_{cat}).

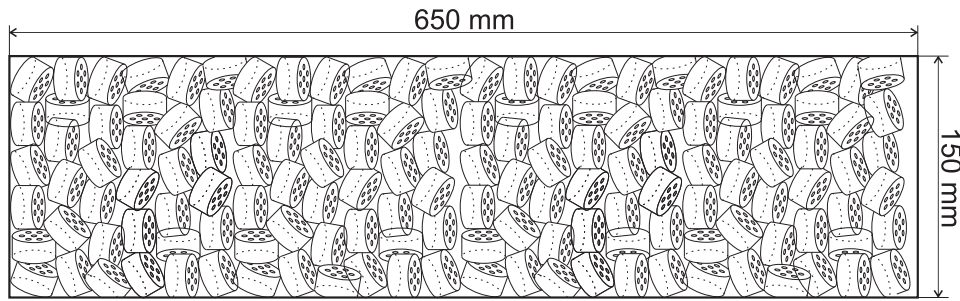


Fig. 6. The computational domain of the reformer.

The standard $k-\epsilon$ model of turbulence is used to simulate fluid dynamics. $k-\epsilon$ model is the most common model used in Computational Fluid Dynamics (CFD) to simulate the flow characteristics for flow conditions. The theory and mathematical formulation of the $k-\epsilon$ model is well described in ANSYS Fluent Theory Guide [56,57].

3.2. Kinetic model

The calculations of reaction rate for SMR process in present investigation are based on the nuanced analysis and investigations of chemical reactions that are presented in scientific publication [2,21]. It has been argued that both CH_4 and H_2O were adsorbed on the catalyst layer with dissociation. The expressions for determination of reaction rate suggested by Froment and Xu [2] are employed in this model.

The reaction rate of the methane consumption through reactions (1) and (3) can be found from following expression:

$$r_{\text{CH}_4} = R_1 + R_3. \quad (11)$$

The reaction rate of carbon monoxide (CO) and carbon dioxide (CO_2) production can be found from following expressions:

$$r_{\text{CO}} = R_1 - R_2. \quad (12)$$

$$r_{\text{CO}_2} = R_3 + R_2. \quad (13)$$

The rate of each reaction can be determined from following expressions [21]:

$$R_1 = \frac{k_1}{p_{\text{H}_2}^{2.5}} \left(p_{\text{CH}_4} \cdot p_{\text{H}_2\text{O}} - \frac{p_{\text{H}_2}^3 \cdot p_{\text{CO}}}{K_{e1}} \right) \cdot \frac{1}{Q_r^2}; \quad (14)$$

$$R_2 = \frac{k_2}{p_{\text{H}_2}} \left(p_{\text{CO}} \cdot p_{\text{H}_2\text{O}} - \frac{p_{\text{H}_2} \cdot p_{\text{CO}_2}}{K_{e2}} \right) \cdot \frac{1}{Q_r^2}; \quad (15)$$

$$R_3 = \frac{k_3}{p_{\text{H}_2}^{3.5}} \left(p_{\text{CH}_4} \cdot p_{\text{H}_2\text{O}}^2 - \frac{p_{\text{H}_2}^4 \cdot p_{\text{CO}_2}}{K_{e3}} \right) \cdot \frac{1}{Q_r^2}; \quad (16)$$

$$Q_r = 1 + K_{\text{CO}} p_{\text{CO}} + K_{\text{H}_2} p_{\text{H}_2} + K_{\text{CH}_4} p_{\text{CH}_4} + \frac{K_{\text{H}_2\text{O}} p_{\text{H}_2\text{O}}}{p_{\text{H}_2}}, \quad (17)$$

where p_i – partial pressure of i reacting element, bar; K_{je} – equilibrium constant of (1)–(3) reactions, respectively; k_j – kinetic rate constant of j reaction.

In Eq. (17) adsorption constants of i element (K_i) can be determined as:

$$K_i = K_{0i} \cdot e^{-\Delta H_i / (R_0 T)}, \quad (18)$$

where H_i – adsorption enthalpy of i element, J/mol; K_{0i} – constant of i component.

3.3. Boundary conditions

The boundary and initial conditions for the computational domain of the reformer shown in Fig. 6 are determined as follows:

– reformer inlet:

$$x = 0; \quad T_g = T_{g(in)}; \quad C_i = C_{i(in)}. \quad (19)$$

– reformer outlet:

$$x = L; \quad \frac{\partial C_i}{\partial x} = 0. \quad (20)$$

– reformer center:

$$y = 0; \quad \frac{\partial C_i}{\partial y} = 0; \quad \frac{\partial T_g}{\partial y} = 0. \quad (21)$$

The properties of reacting flow in the steam methane reformer (for example, density and specific heat) are as function of temperature and gas flow composition. These properties are calculated by rule of mixtures from the properties and mass or mole fractions of the each element of the reaction flow. The properties of individual gas components are taken from Fluent Database [56].

4. Results and discussion

The equations of CFD model were solved via ANSYS Fluent 16.2. Solutions are converged approximately at 1200 iterations for all investigated initial conditions.

4.1. Pressure drop

To validate developed CFD model, pressure drop determined via CFD model was compared with experimental results obtained on experimental setup that is described above. Fig. 7 shows the dependence of pressure loss on the gas flow velocity at the inlet for pressure $p = 1$ bar and gas temperature of 450° . According depicted results, agreement between the calculated and experimental results are obvious. In addition, a good correlation of the results is observed for other temperatures. The author carried out studies for 50° , 200° , 300° and 400° . For all cases, the correlation was obvious. To simplify this article, these graphs were not shown. Discrepancies between the data are insignificant. These discrepancies can be explained by errors of measurement unit of experimental setup and assumptions for computational fluid dynamics model. Therefore, well agreement of the results allows to state that the selected model setup of Fluent solver are correct for modeling of fluid dynamics. Fig. 7 depicts that an increase of the gas flow rate leads to pressure loss increasing.

4.2. Mass fraction species and temperature distributions

The model results of temperature and mole fraction of individual element distribution in steam methane reformer presented in Fig. 8. The reaction mixture here runs from bottom to up. Also here x is axial direction, y is radial direction. As mentioned above, the reformer has a

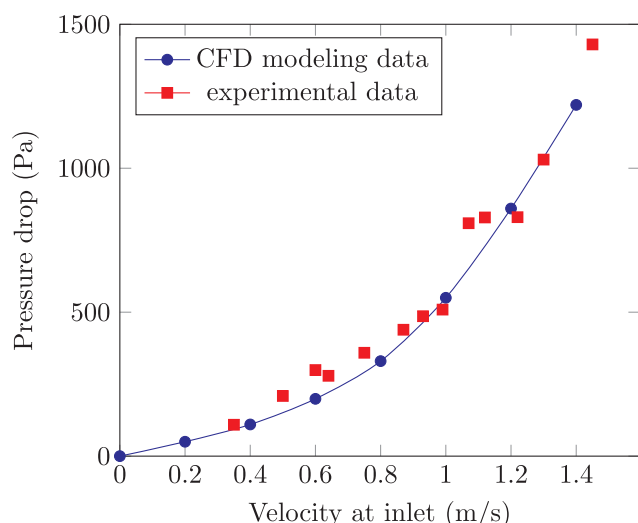


Fig. 7. Comparison of pressure drop obtained from CFD modeling and experiment.

constant catalyst layer porosity in x (axial) and y (radial) directions, and reformer wall is thermally perfect, no heat losses to surroundings. For a better analysis of the results, the scale along the y -axis is reduced, because in the reactor there is no significant the flow properties gradients along the y -axis.

The distribution of reaction mixture temperature and mole fractions of chemical elements are obtained at follows initial conditions. Before steam methane reforming the temperature is constant and uniform for catalyst bed ($T = 1300$ K). Steam-to-methane ratio is 2.0. Pressure is 5 bar. Residence time is $4.54 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$. Temperature of feed stock is 800 K. Other catalyst properties are presented in Table 1.

Along the length of reformer the mole fraction of methane and steam decreases, while the component concentration of synthesis gas such as hydrogen and carbon monoxide increase. The mole fraction of CH_4 is constantly decreasing due to taking place of reaction (1) and (3). However near reformer outlet the mole fraction is little changed, because the catalyst temperature is not enough for steam methane reforming reactions and the mole fraction of methane is small. On the other hand, the mole fraction of H_2O decreases constantly from the inlet

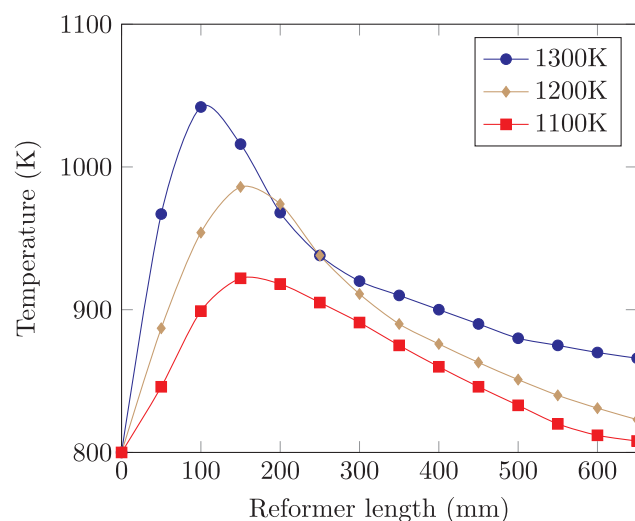


Fig. 9. Distribution of reaction mixture temperature along reformer length for various catalyst temperature.

to the outlet of reformer. This is due to the fact that steam is consumed not only by steam methane reactions but also by the water gas-shift reaction.

The mole fraction of hydrogen and carbon monoxide increases continuously from the inlet to the outlet of reformer. However, mole fraction gradient of the carbon monoxide decreases near the reformer outlet, because some of the carbon monoxide is consumed by the water gas-shift reaction.

The temperature gradient near the reformer inlet has a maximum value, because the rate of reactions (1) and (3) is maximum and the heat transfer between catalyst and reaction flow are strong in this section of the reformer. The temperature gradient slows down to outlet of reformer, because the catalyst temperature decreases due to endothermic steam methane reforming reaction, and also due to increase of steam-to-methane ratio from inlet to outlet due to methane consumption. The temperature gradient has no noticeable changes in the radial (y) direction of the reactor, because catalyst layer porosity is constant and no heat losses through the wall to surroundings.

The temperature distributions for various initial catalysts temperatures are presented in Fig. 9. The initial temperature of inlet gas mixture

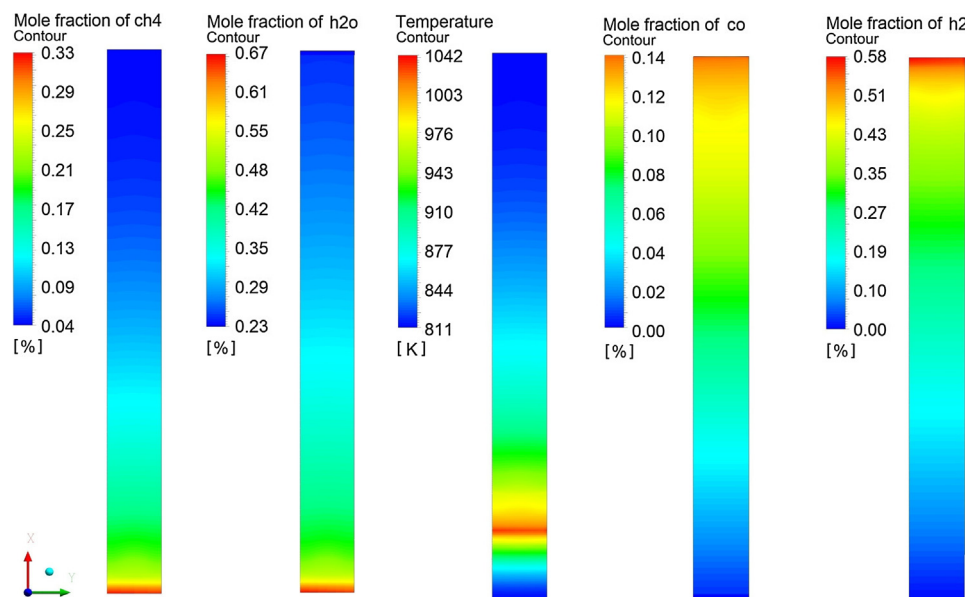


Fig. 8. Distribution of temperature and mole fraction of individual element in the reformer.

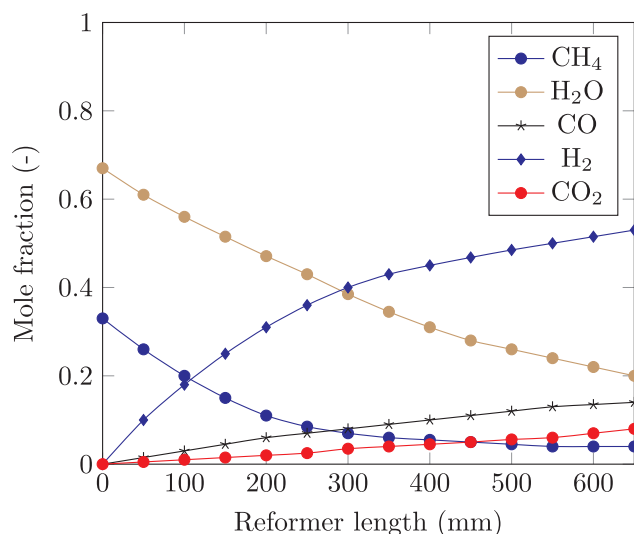


Fig. 10. Distribution of mole fraction along reformer length at steam-to-methane ratio 2, pressure 5 bar, residence time $4.54 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$.

is 800 K for all cases. Steam-to-methane ratio is 2.0. Pressure is 5 bar. As is seen from Fig. 9 in the section of reformer near inlet the reaction mixture temperature increases due to heat transfer between catalysts and reaction flow. The maximum gas temperature is reached at a distance of 100–200 mm from the inlet. After reaching the maximum value, the gas temperature is gradually decreasing due to the endothermic reaction of steam methane reforming.

The developed CFD model allows to calculating mole fraction of each individual element at any point of steam methane reformer. Fig. 10 shows the distribution of mole fraction CH_4 , H_2O , CO , CO_2 and H_2 along reformer length for initial conditions which is mentioned above at the catalyst temperature is 1200 K. As can be seen the mole fractions of each element are constantly changing due to chemical reactions when the reaction mixture passes through catalyst bed of the steam methane reformer.

4.3. CH_4 conversion rate and H_2 yield

For thermochemical waste-heat recovery systems and for energy efficiency of overall steam methane reforming process the determination of the methane conversion rate is significant importance. By the first law of thermodynamics, the amount of energy entering any reformer at steady state must equal the energy leaving that reformer [58]. Put it another way, the quantity of heat that is converted to the chemical energy of a syngas is determined by the enthalpy of the steam methane reforming reaction [34]. On the other hand, the enthalpy of the steam methane reforming reaction is determined by the value of methane conversion rate [59,60].

The methane (CH_4) conversion rate and hydrogen (H_2) yield are determined as follows:

$$\chi_{\text{CH}_4} = \frac{[\text{CH}_4]_{\text{in}} - [\text{CH}_4]_{\text{out}}}{[\text{CH}_4]_{\text{in}}}; \quad (22)$$

$$y_{\text{H}_2} = \frac{[\text{H}_2]_{\text{out}}}{[\text{CH}_4]_{\text{in}}}; \quad (23)$$

Fig. 11 shows the overall methane conversion rate as a function of residence time (catalysts layer mass per mole flowrate on methane inlet) for steam-to-methane ratio 2 and for various catalyst temperatures. For all cases the inlet gas temperature is 800 K. As can be seen in Fig. 11 the methane conversion rate increases with increasing residence time. The methane conversion rate decreases with decreasing catalyst temperature, because the heat that is contained in the catalyst bed is

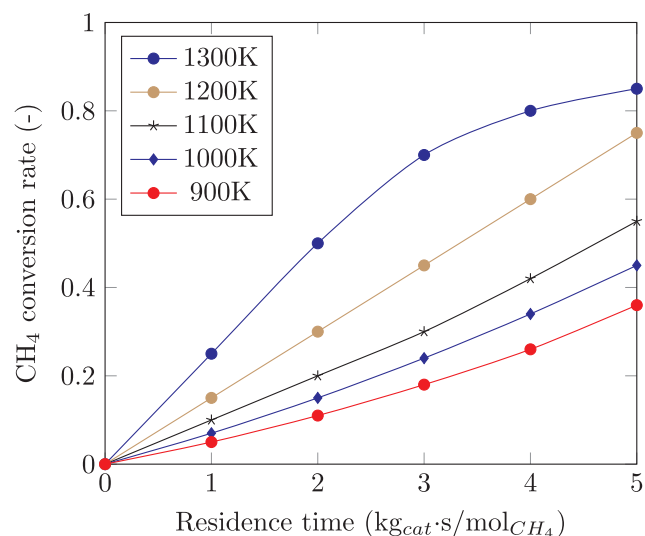


Fig. 11. Methane conversion against residence time at different catalyst temperature for steam-to-methane ratio 2.

insufficient to fully complete the steam methane reforming reaction. Piece of methane is unreacted. Further analysis showed that as the temperature of the catalyst increases above 1300 K, the SMR reaction can reach a state that is close to equilibrium.

Fig. 12 shows the variation of hydrogen yield against residence time at different catalyst temperature for steam-to-methane ratio 2. It shows that the hydrogen formations increase with increasing the catalyst temperature. The yield of hydrogen is constantly increasing with increasing residence time. This fact well agrees with Fig. 11 showing that methane conversion rate is increasing with increasing residence time. Therefore, the yield of hydrogen produced by reactions ((1)–(3)) increases, and at a catalyst temperature above 1300 K for residence time above $5 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$ it reaches close to an equilibrium value.

The comparison of synthesis gas composition is presented in Table 2. There are two comparable results. First is equilibrium syngas composition calculated by Vekhviker and Kravchenko [52] for the methane steam reforming at steam-to-methane ratio 2. Second is syngas composition obtained by presented CFD model at residence time $4.54 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$ and steam-to-methane ratio 2.0.

As can be seen from Table 2, for the temperature 1000 K there is a

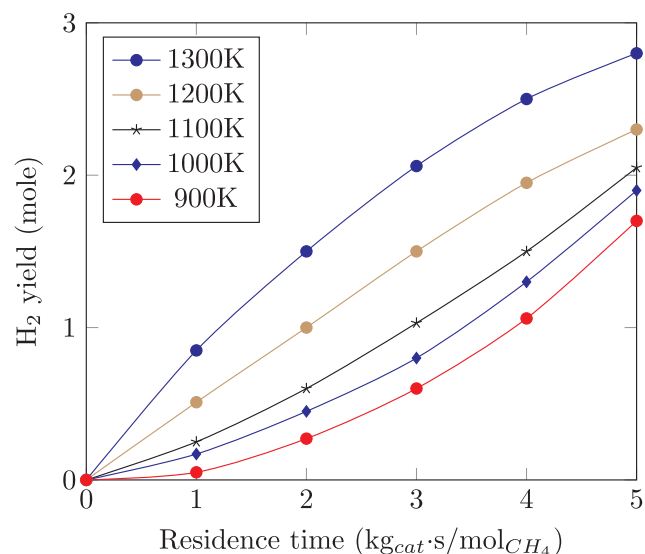


Fig. 12. Hydrogen yield against residence time at different catalyst temperature for steam-to-methane ratio 2.

Table 2

Comparison of synthesis gas composition, % (mol/mol).

Temperature	Pressure	Source	Gas composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000 K	1 bar	Equilibrium	63.42	14.43	16.10	5.12	0.92
		CFD	53.84	8.15	23.16	5.88	8.97
1000 K	5 bar	Equilibrium	61.44	14.93	17.97	4.18	1.48
		CFD	50.12	7.83	26.07	5.31	10.67
1100 K	1 bar	Equilibrium	63.76	16.11	16.11	3.94	0.08
		CFD	56.09	12.65	20.60	4.72	5.94
1100 K	5 bar	Equilibrium	61.45	14.93	17.96	4.18	1.48
		CFD	55.32	10.99	22.57	4.99	6.13

large discrepancy between the compositions of synthesis gas. However, with increasing temperature, this discrepancy decreases and at a temperature above 1300 K the synthesis gas composition obtained by CFD model is close to the equilibrium composition.

5. Conclusion

1. In this paper, the CFD-modeling of steam methane reforming process over a preheated Ni-based catalyst is reported. To validate the solver setup of the fluid dynamic for developed CFD model, pressure drops are compared with experimental results. It was observed that each 100 mm of catalyst bed gives pressure drop about 160 Pa for gauge pressure of 1 bar. The correlation between the calculated and experimental results are obvious.
2. The developed CFD model allows to calculate mole fraction of each individual element at any point of steam methane reformer. The mole fractions of each element are constantly changing due to chemical reactions when the reaction mixture passes through catalyst bed of the steam methane reformer. In the reformer there is no significant the flow properties gradients along the radial axis. Comparing the synthesis gas composition obtained with the CFD model versus the equilibrium syngas composition, it can be seen that at the catalyst temperature above 1300 K, a syngas composition close to equilibrium is achieved. The maximum temperature about of reaction gas mixture is observed at initial part of catalyst bed (100–200 mm from inlet).
3. In the analyzed way of steam methane reforming, the heat for the endothermic reaction comes from the pre-heated catalyst; therefore the catalyst temperature along the reformer is variable. The maximum temperature gradient is observed at the section near the reformer inlet, because this section has the maximum rate of steam methane reforming reaction and the maximum temperature drop between catalysts and reaction flow. For follow initial conditions (steam-to-methane ratio is 2.0, pressure is 5 bar, residence time is $4.54 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$, temperature of feed stock 800 K, catalyst temperature 1300 K), temperature of outlet synthesis gas is 849 K.

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